

七. 凝聚理论与合作现象

1. 有限体积系统

$$V \rightarrow \infty: \underbrace{\frac{P}{kT} = \sum \frac{1}{\lambda^3} \frac{Z_i^1}{V}}_{F(z) \rightarrow z=1, F(z) \text{ 奇异}} \cdot \rho = \sum \frac{1}{\lambda^3} \frac{Z_i^1}{V} \cdot \lambda = \sqrt{\frac{2\pi}{mkT}} \hbar$$

迈耶猜测: 非理想气体平衡时满足: $(\frac{P}{kT})_V = \sum b_i(V) y^i$, $(\rho)_V = \sum b_i(V) y^i$ ($y = \frac{e^{\frac{\mu}{kT}}}{\lambda^3}$)

体积有限/无限情况不同 李政道讲之 $\hookrightarrow$ 延拓到复平面(y): $(\frac{P}{kT})_V = \sum b_i y^i = F(y)$, $\rho_V = y \frac{dF}{dy}$

对有限体积:

硬球模型: $u_{ij} = \begin{cases} \infty & r_{ij} \leq a \\ 0 & r_{ij} > b \\ \geq u_0 & \text{else} \end{cases} \Rightarrow$

有限V下 $N_0 \leq \frac{V}{\frac{4}{3}\pi a^3}$, 粒子数有限 $\Rightarrow Q_V = \sum_{N=0}^{N_0} \frac{y^N}{N!} Q_V(N) \Rightarrow N_0$ 个复根, $Q_V(N) > 0$
 $= 1 + Vy + \dots + \frac{Q_V(N_0)}{N_0!} y^{N_0}$

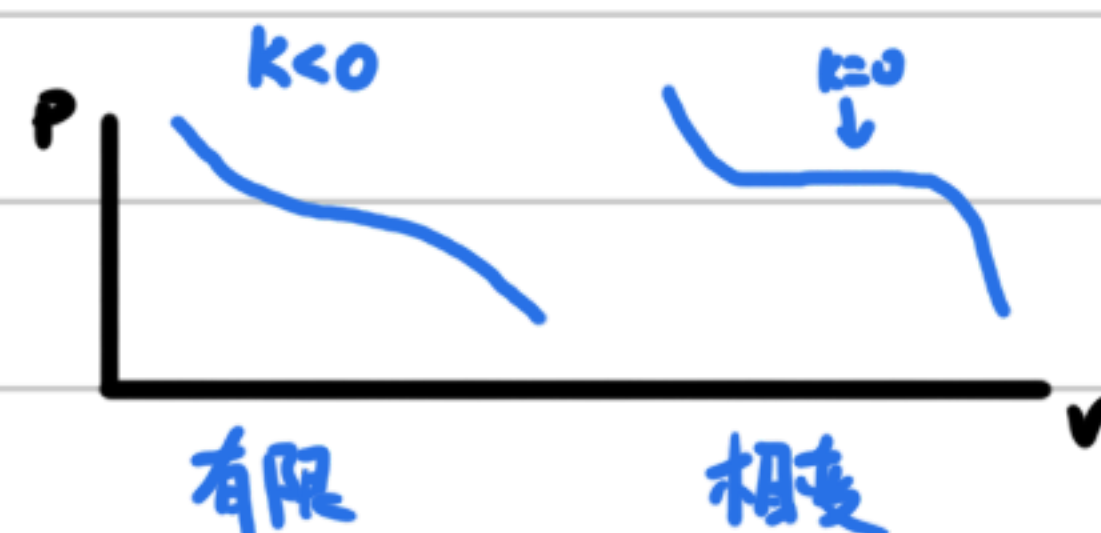
多项式分解: $Q_V = \prod_{i=1}^{N_0} (1 - \frac{y}{y_i})$

$Q_V(N)$ 为正 $\Rightarrow y > 0$ 无正根 $\Rightarrow 1) y_i \notin \mathbb{R}^+$

2) 物理区域内 $\frac{P}{kT} = \ln Q$ 解析, $P = \frac{\partial}{\partial \ln y} \ln Q$ 解析

$$\frac{P}{kT} = \frac{1}{V} \sum_{i=1}^{N_0} \ln(1 - \frac{y}{y_i}) \cdot P = \frac{1}{V} \sum_{i=1}^{N_0} \frac{1}{1 - \frac{y}{y_i}} (-\frac{y}{y_i})$$

3) 比容 $v \equiv \frac{1}{\rho}$, 压强P为 v, T 解析 $P(v, T)$



有限体积不会发生相变

有限体积不会发生相变 $\leftarrow$

- $P > 0$ 有限 $\frac{1}{V} \frac{y}{Q_V} \frac{\partial Q_V}{\partial y} = \frac{1}{Q_V} \sum y^N \frac{Q_V(N)}{N!} \frac{N}{V} = \frac{\langle N \rangle}{V}$
- $(\frac{\partial P}{\partial \ln y})_T > 0$ 有限 $\frac{y}{V} \frac{\partial P}{\partial y} = \frac{1}{V} (\frac{1}{Q_V} \frac{\partial Q_V}{\partial \ln y} - \frac{1}{Q_V^2} (\frac{\partial Q_V}{\partial \ln y})^2)$
 $= \frac{1}{V} (\langle N^2 \rangle - \langle N \rangle^2) = \frac{\langle (N - \langle N \rangle)^2 \rangle}{V}$
- $(\frac{\partial P}{\partial v})_T > 0$ 有限 $\frac{kT \frac{\partial}{\partial \ln y} (\frac{P}{kT})}{\frac{\partial P}{\partial \ln y}} = \frac{kT P}{\frac{\partial P}{\partial \ln y}} = \frac{1}{V \langle (N - \langle N \rangle)^2 \rangle} > 0$

2. 无限容积情况 (李-杨定理)

定理一: $y \in \mathbb{R}^+$ 时, 物理区域内 $\lim_{V \rightarrow \infty} \frac{1}{V} \ln Q = \frac{P}{kT}$ 存在, 为 y 的单增函数

Pf: 分割 $V = V_1 + V_2 + S_b$, b 为粒子有效作用距离 (使 V_1, V_2 不相互作用)



1) 先证 $Q_V > Q_{V_1} \cdot Q_{V_2}$: $Q_{V_1} = \sum \frac{y^{N_1}}{N_1!} Q_{V_1}(N_1)$, 同 V_2, V , $Q_{V_1+V_2} = \sum \frac{y^N}{N!} Q_{V_1+V_2}(N) = Q_{V_1} \cdot Q_{V_2}$ ($Q_{V_1+V_2}(N) = \sum Q_{V_1}(n) \cdot Q_{V_2}(N-n) C_N^n$)

$Q_V > Q_{V_1+V_2} = Q_{V_1} \cdot Q_{V_2}$, done

N_1	n	N_2
V_1	S_b	V_2
a_i	r_k	β_j
a_i'	r_k'	β_j'

2) 若1成立, 则有与 V_1, V_2, S 无关的 K , 使 $Q_V < Q_{V_1} Q_{V_2} K^S$:

考虑左组态, 粒子数 $N = N_1 + N_2 + n$, V_1 粒子名 $\alpha_1, \dots$, V_2 为 β , S_b 内为 γ

$a \neq 0$, b 有限: 每个粒子相互作用粒子数有限: $m < (\frac{b}{a})^3$

不等式: $U = \sum U_{ij} > \sum U_{a_i a_i'} + \sum U_{\beta_j \beta_j'} + \sum U_{\gamma_k \gamma_k'} + nm u_0$ ($u_0 < 0$)

τ 为粒子所占容积, 则 S_b 内积分 $\int_{S_b} e^{-\sum U_{\gamma_k \gamma_k'} / kT} \prod_{k=1}^n d^3 r_k \leq (S_b - n\tau + \tau)(S_b - n\tau + 2\tau) \dots (S_b - \tau)(S_b)$
 $\leq \tau^n \frac{M!}{(M-n)!}$, $\tau = \frac{4\pi}{3} a^3$, $M = \lfloor \frac{S_b}{\tau} \rfloor$

故 $Q_V(N) < \sum \frac{N!}{N_1! N_2! n!} Q_{V_1}(N_1) Q_{V_2}(N_2) \cdot \tau^n \frac{M!}{(M-n)!} e^{-nm u_0 / kT}$ $\leftarrow S_b$ 内粒子全相互作用情况

即: $Q_V < \sum_{N_1, N_2} \frac{y^{N_1+N_2}}{N_1! N_2!} Q_{V_1}(N_1) Q_{V_2}(N_2) \sum_{n=0}^M \frac{M!}{n! (M-n)!} (y \tau e^{-nm u_0 / kT})^n$

$Q_{V_1} Q_{V_2} < Q_V < Q_{V_1} Q_{V_2} \cdot K^S$

$< Q_{V_1} Q_{V_2} \cdot (1 + y \tau e^{-nm u_0 / kT})^M$, 故 $K = (1 + y \tau e^{-nm u_0 / kT})^{M/\tau}$, 与 V_1, V_2, S 无关

3) 推广, 1 个小块堆成大块, $V = V_1 + \dots + V_i + S_b$

$\prod_{i=1}^i Q_{V_i} < Q_V < K^S \prod_{i=1}^i Q_{V_i}$

4) 取 V_n 为立方体 $(2^n L)^3$, 内有 $V_n' \Rightarrow V_n = V_n' + S_n \cdot b$, $S_n b = 6 \cdot \frac{1}{2} (2^n L)^2$

放大 n 次: $8^n L^3 \Rightarrow (Q_{V_n})^8 \cdot K^{-8^n S_n} < (Q_{V_n})^8 < Q_{V_{n+1}} < (Q_{V_n})^8 \cdot K^{8^n S_n}$

$8^n [\ln Q_{V_n} - S_n \ln K] < \ln Q_{V_{n+1}} < 8^n [\ln Q_{V_n} + S_n \ln K]$, 同除 $V_{n+1} \approx 8^n V_n \Rightarrow \frac{\ln Q_{V_{n+1}}}{V_{n+1}}$ 夹于 $\frac{\ln Q_{V_n}}{V_n} \pm \frac{S_n}{V_n} \ln K$

5) 对任一形状, 同上, 等比放大即可 (不能分形)

综上, 证毕, 当 $N \rightarrow \infty$ 时有 $\frac{P}{kT} \equiv \lim_{N \rightarrow \infty} \frac{\ln Q_N}{V_N}$

6) 证 $\frac{P}{kT}$ 对 y 单调: $\frac{\partial}{\partial y} \frac{1}{V} \ln Q_V = \frac{1}{V} \frac{\partial}{\partial y} \sum N y^N Q_V(N) \frac{1}{N!} = \frac{\langle N \rangle}{V} < \frac{N_0}{V} = \rho_{\max}$

故 $y \neq 0$ 时 $\frac{P}{kT}$ 对 y 单增, 增速有界 (若 $\lim_{y \rightarrow 0} \frac{1}{V} \ln Q_V$ 存在, 则可用同方法证单增, $\frac{\langle N^n \rangle}{V}$)



定理二. y 的复数域上, 若区域 R 不含根, 则 R 上 $\lim_{v \rightarrow \infty} \frac{1}{v} \frac{\partial}{\partial \ln y} \ln Q_v$ 均对 y 解析, 且其 $\lim$ 与 $\frac{\partial}{\partial \ln y}$ 对易

证明: 1) R 包含一段实轴, 故取 $y_0 \in R$ 为实, 作圆 (r, y_0)



$z_i = y_i - y_0$. $\exists$ 开球 $B(d, y_0)$, $\forall y_i \notin B(d, y_0)$ (所有零点不包含)

$$Q_v(y) = \prod_{i=1}^N (1 - \frac{y}{y_i}) = \prod \frac{z_i - z}{y_i} = \prod (1 - \frac{z}{z_i}) (\frac{z_i}{y_i}) \Rightarrow \frac{1}{v} \ln Q_v(y) = \sum_{i=1}^N \frac{1}{v} \ln (1 - \frac{z}{z_i}) + \sum_{i=1}^N \frac{1}{v} \ln \frac{z_i}{y_i} \stackrel{\text{收敛}}{=} \sum_{i=1}^N a_i(v) z^i \equiv S_v(z)$$

$z=0$ 时 $a_0(v) = \frac{1}{v} \ln Q_v(y_0)$ 存在; $l \geq 1$ 时: $\rho_0 = \frac{N_0}{v}$ (最大密度) $a_l(v) = \frac{1}{lv} \sum_{i=1}^N (\frac{1}{z_i})^l$. $|a_l| < \frac{1}{l} \rho_0 (\frac{1}{d})^l \Rightarrow |z| < d$ 时 $\sum a_l z^l$ 一致收敛

2) 证明 $a_l(v)$ 有界, $\lim_{v \rightarrow \infty} a_l(v)$ 存在

(定理一) $\frac{1}{v} \ln Q_v(y)$ 存在, 即 $\lim_{v \rightarrow \infty} S_v(z)$ 存在, $S_v(0) = a_0(v)$ 存在 $\Rightarrow$ 收敛速速: $\forall \epsilon, \exists v_0$ 使 $\forall v, v' > v_0$

$$1. |\sum_{i=1}^N a_i(v) \epsilon^i| < |\sum_{i=1}^N \rho_0 (\frac{\epsilon}{d})^i| = \frac{\rho_0 (\frac{\epsilon}{d})^2}{1 - (\frac{\epsilon}{d})^2}$$

$$|S_v(z) - S_{v'}(z)| < \epsilon^2, |a_0(v) - a_0(v')| < \epsilon^2$$

$$2. \epsilon a_l(v) = \lim_{v \rightarrow \infty} (S_v(z) - a_0 - \sum_{i=1}^{l-1} a_i(v) \epsilon^i). \text{ 故 } \epsilon |a_l(v) - a_l(v')| < \text{常} \cdot \epsilon^2 \Rightarrow \lim_{v \rightarrow \infty} a_l(v) \text{ 存在}$$

3. 同理, 取 ϵ^i 为 ϵ^{m+i} , 等号 $\sum_{i=1}^{\infty}$ 可证 $a_n(v)$ 存在

$$\text{故 } \lim_{v \rightarrow \infty} a_l(v) \text{ 存在, } \lim_{v \rightarrow \infty} \sum a_l(v) z^l = \sum a_l(\infty) z^l$$

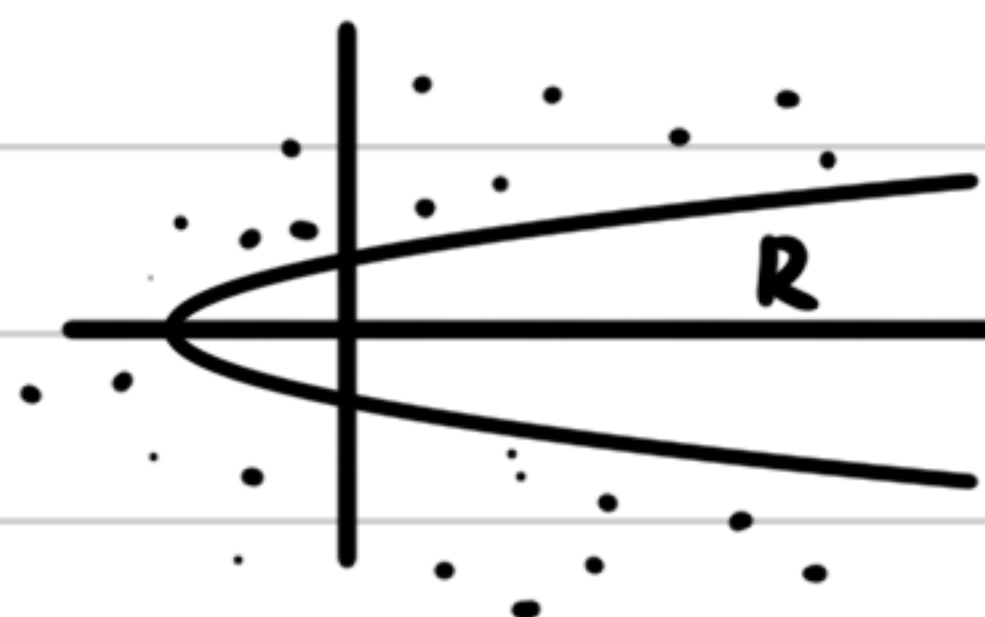
$$3) \text{ 证明 } \frac{\partial}{\partial \ln y} \text{ 与 } \lim \text{ 对易: } z \frac{\partial}{\partial z} \sum a_l(v) z^l = \sum a_l(v) l z^l, |a_l(\infty)| < \frac{\rho_0}{l} (\frac{1}{d})^l \Rightarrow |l a_l(\infty)| < \rho_0 (\frac{1}{d})^l$$

$$\text{故 } z \frac{\partial}{\partial z} \lim \sum a_l(v) z^l = z \frac{\partial}{\partial z} \sum a_l(\infty) z^l = \sum a_l(\infty) l z^l$$

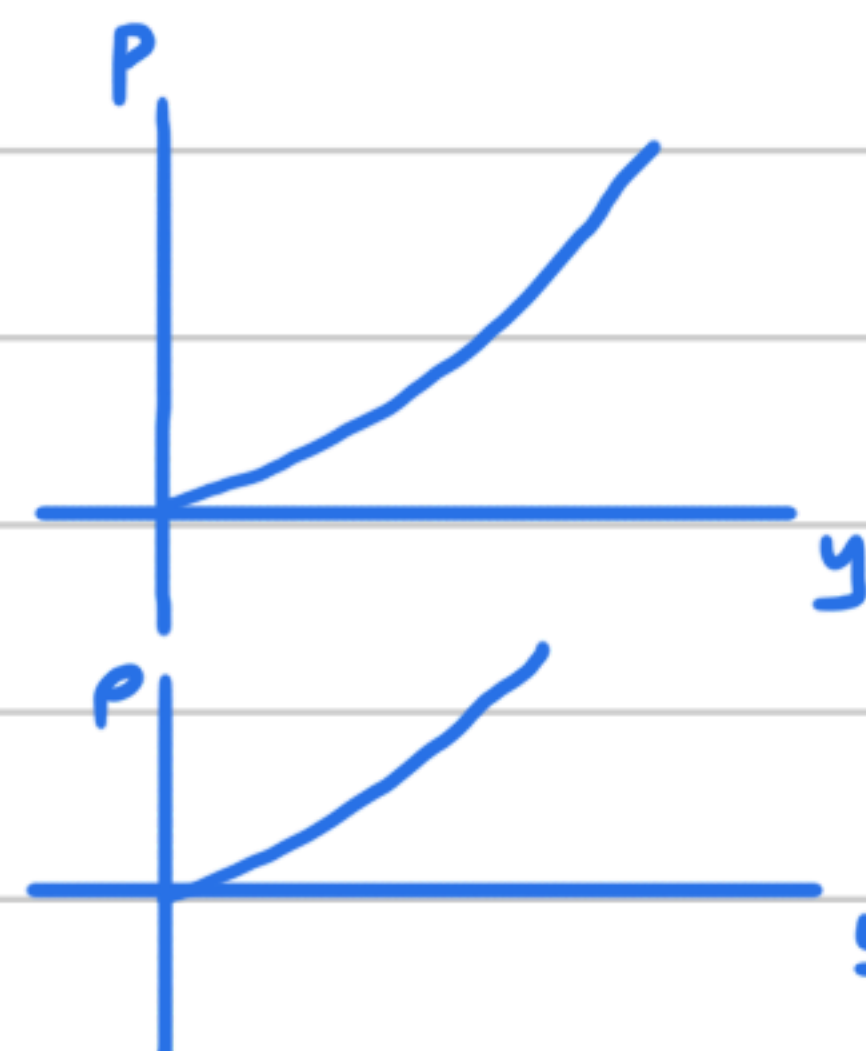
$$\lim z \frac{\partial}{\partial z} \sum a_l(v) z^l = \lim \sum l a_l(v) z^l = \sum l a_l(\infty) z^l$$

相变的李杨定理:

1. 无根区:

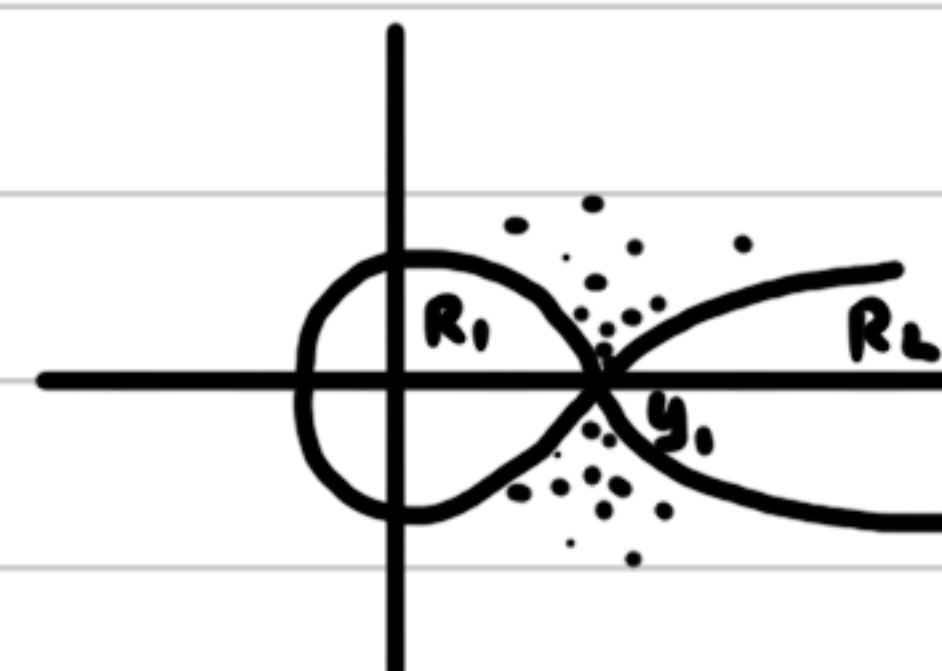


$\Rightarrow$

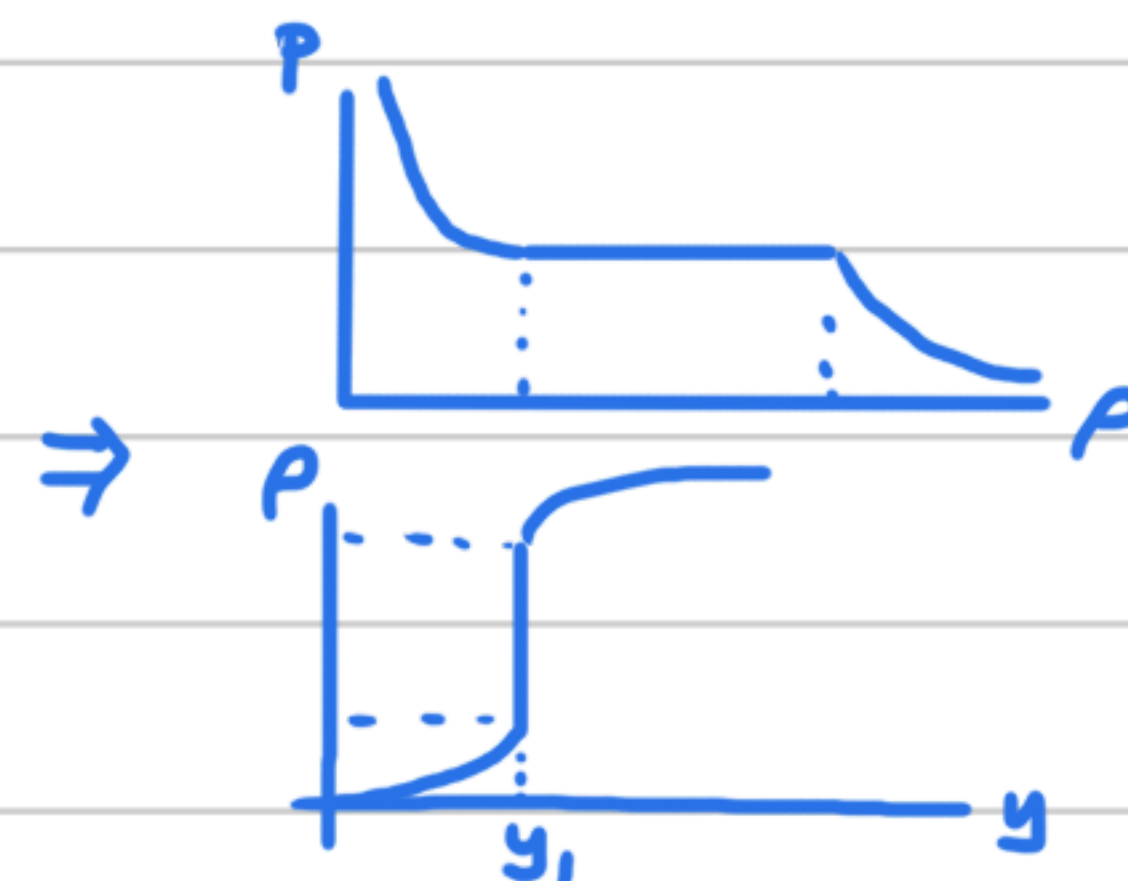


R 内定理 1, 2 都满足

$$\rho = \frac{\partial}{\partial \ln y} (\frac{P}{kT}) \text{ 存在, } \frac{\partial P}{\partial \rho} = \frac{\partial P / \partial \ln y}{\partial \rho / \partial \ln y} \text{ 不散 } (\frac{\partial P}{\partial \ln y} \text{ 存在, 故 } \frac{\partial P}{\partial \ln y} > \frac{\langle N_m \rangle}{V} = 0)$$



y_1 点, $\frac{1}{v} \ln Q_v$ 单增, 但 $\lim_{v \rightarrow \infty} (\frac{1}{v} \ln Q_v)$ 不存
故 P 随 y 连续, R_1 与 R_2 处 P 随 y 连续, 无穷远
 y_1 点处: P 可能不解析 $\Rightarrow$ 存在跳变



2. 零点、密集度:

密集度大: -阶不连续 $\rightarrow$ -级相变

较小: 前 $n-1$ 阶连续, n 阶不连续 $\rightarrow n$ 级相变 ? 数学语言?

3. 例: Ising 模型相变

Ising 模型与格点气体:

有序-无序模型: A, B 原子在格点

$$U_{0-D} = N_{AA} \epsilon_{AA} + N_{BB} \epsilon_{BB} + N_{AB} \epsilon_{AB}$$

$$Q_{0-D} = \sum_{\{i\}} e^{-\frac{U}{kT}} = e^{-\frac{F}{kT}}$$

N_{AA} : A-A 近邻数, N_{BB} , N_{AB} 同

关系: $2N_{AA} + N_{AB} = nN_A$, 同 B; $N_A + N_B = N$

$$\text{故 } U_0 = N_{AA}(\epsilon_{AA} + \epsilon_{BB} - 2\epsilon_{AB}) + nN_A(\epsilon_{AB} - \epsilon_{BB}) + N \frac{n}{2} \epsilon_{BB}$$

Ising:

$$\text{势能: } U_{Ising} = (N_{\uparrow\uparrow} + N_{\downarrow\downarrow}) \epsilon_{\uparrow\uparrow} + N_{\uparrow\downarrow} \epsilon_{\uparrow\downarrow} + \mu H (N_{\uparrow} - N_{\downarrow})$$

$$\text{则配分 } Q = \sum_{\{i\}} e^{-\frac{U}{kT}} = e^{-\frac{F}{kT}} \quad (F = kT \ln Q)$$

$N_{\uparrow/\downarrow}$ 自旋向上/下 粒数, n 为近邻数

$N_{\uparrow\uparrow}$ 相邻子取向为 $\uparrow$ 的对数, H 为 z 向磁场

$$2N_{\uparrow\uparrow} + N_{\uparrow\downarrow} = nN_{\uparrow}, \text{ 同 } \downarrow$$

$$\text{故 } N_{\downarrow\downarrow} = \frac{1}{2}(nN_{\downarrow} - N_{\uparrow\downarrow}) = \frac{1}{2}(n(N - N_{\uparrow}) + 2N_{\uparrow\uparrow} - nN_{\uparrow})$$

$$= \frac{1}{2}nN - nN_{\uparrow} + N_{\uparrow\uparrow},$$

$$U = N_{\uparrow\uparrow}(2\epsilon_{\uparrow\uparrow} - 2\epsilon_{\uparrow\downarrow}) + N_{\uparrow\downarrow}[-n\epsilon_{\uparrow\uparrow} + n\epsilon_{\downarrow\downarrow} - 2\mu H] + N(\mu H + \frac{n}{2}\epsilon_{\uparrow\uparrow})$$

格气: N 个不可分辨粒子分布 N 个格点

$$Q_N = \frac{1}{(2\pi)^{3N}} \prod_i d^3 p_i e^{-\frac{p_i^2}{2mT}} \cdot \prod_i d^3 r_i e^{-U/kT} \cong \frac{1}{\lambda^{3N}} \sum e^{-\frac{U}{kT}} \quad (\lambda = \sqrt{\frac{2\pi}{mT}} \text{ 为热波长, 为动量空间积分})$$

$$\text{邻近势能 } U_{ij} = \begin{cases} u & r=0 \\ 0 & \text{else} \end{cases} \Rightarrow U = \sum U_{ij} \Rightarrow Q_g = \sum \frac{u^N}{N!} Q_v(N), \quad u = \frac{u}{\lambda^3}$$

$$\text{故 } U_g = \sum U_{ij} = u \cdot N_{..},$$

三者等价:

格气

u

u

u^N

$$Q = \sum \frac{u^N}{N!} Q_v(N) = \sum \frac{u^N}{N!} e^{-\frac{U}{kT}}$$

$$P = \frac{kT}{N} \ln Q = \frac{kT}{N} \ln Q_{v(N=1)}$$

$2P-1$

P

Ising

$$2(\epsilon_{\uparrow\uparrow} - \epsilon_{\uparrow\downarrow}) = 4\epsilon$$

$$\frac{2\mu H + n\epsilon}{kT} (\epsilon_{\uparrow\uparrow} - \epsilon_{\uparrow\downarrow})$$

$$\frac{2(\mu H + n\epsilon)}{kT} N_{\uparrow}$$

$$Q = e^{\frac{N}{kT}(\mu H + \frac{1}{2}\epsilon_{\uparrow\uparrow})} = \sum e^{-\frac{U_i}{kT}} e^{\frac{N}{kT}(\mu H + \frac{1}{2}\epsilon_{\uparrow\uparrow})}$$

$$\frac{kT}{N} \ln Q = -F + \mu H + \frac{1}{2}n\epsilon_{\uparrow\uparrow}$$

$$M \equiv \frac{\langle N_{\uparrow} - N_{\downarrow} \rangle}{N} = \frac{\langle 2N_{\uparrow} - N \rangle}{N}$$

$$\frac{M+1}{2}$$

O-D

$$\epsilon_{\uparrow\uparrow} + \epsilon_{\downarrow\downarrow} - 2\epsilon_{\uparrow\downarrow}$$

$$\frac{n(\epsilon_{\uparrow\uparrow} - \epsilon_{\uparrow\downarrow})}{kT}$$

$$\frac{n(\epsilon_{\uparrow\uparrow} - \epsilon_{\uparrow\downarrow})}{kT} N_{AA}$$

$$Q_{0-D} e^{\frac{N n \epsilon_{\uparrow\uparrow}}{kT}}$$

待

故 Ising 模型 $Q = \sum e^{-\frac{U}{kT}}$